# assignment _newton

This is a preview of the published version of the quiz

Started: Apr 29 at 7:59pm

# Quiz Instructions

Remember the simulation in class. NO TUTORIAL

Fnet = 0 -> the object is at rest or moving at a constant velocity : like pushing a sofa at a constant speed since push = friction or a parachute reaching terminal speed since drag = weight.

Fnet = constant ->there is an acceleration. the object accelerates at a constant rate.

Newton's second law -> Fnet = mass x acceleration Use components of forces along axis.

Question 1 10 pts

Whenever the net force on an object is zero, its acceleration

- may be more than zero.
- is zero.
- may be less than zero.

Question 2 10 pts

If an apple experiences a constant net force, it will have a constant

- position.
- acceleration.
- more than one of the above
- speed.
- velocity.

Question 3 10 pts

A mobile phone is pulled northward by a force of 10 N and at the same time pulled southward by another force of 15 N. The resultant force on the phone is

5 N.

0 N.

150 N.

25 N.

Question 4 10 pts

The force of friction on a sliding object is 10 N. The applied force needed to maintain a constant velocity is

more than 10 N.

10 N.

less than 10 N.

Question 5 10 pts

A 10-N falling object encounters 4 N of air resistance. The net force on the object is

0 N.

none of the above

10 N.

4 N.

6 N.

Question 6 10 pts

The force required to maintain a constant velocity for an astronaut in free space is equal to

(remember law of inertia. first laww. see slides again)

zero.

the mass of the astronaut.

- the force required to stop the astronaut.
- none of the above
- the weight of the astronaut.

Question 7 10 pts

A push on a 1-kg brick accelerates it. Neglecting friction, equally accelerating a 10-kg brick requires

- one-tenth the amount of force.
- 10 times as much force.
- just as much force.
- 100 times as much force.
- none of the above

Question 8 10 pts

A jumbo jet has a mass of 100,000 kg. The thrust for each of its four engines is 50,000 N. What is the jet's acceleration when taking off?

- 2 $m/s^2$
- none of the above
- 0.25 $m/s^2$
- 4 $m/s^2$
- 1 $m/s^2$

Question 9 10 pts

When a falling object has reached its terminal velocity, its acceleration is

- *g*.
- zero.

constant.

Question 10 10 pts

A 500-N parachutist opens his chute and experiences an air resistance force of 800 N. The net force on the parachutist is then

500 N upward.

300 N downward.

800 N downward.

500 N downward.

300 N upward.

Question 11 4 pts

Arnold Strongman and Suzie Small each pull very hard on opposite ends of a rope in a tug-of-war. The greater force on the rope is exerted by

(newton's third law. See lecture, slides and TEXTBOOK!)

It depends on how muscular is Suzie

Arnold, of course.

Suzie, surprisingly.

both the same, interestingly

Question 12 6 pts

You push a crate of 100kg and the kinetic firction is 188N. How hard you need to push to get an acceleration of 1m/s/s

2880N

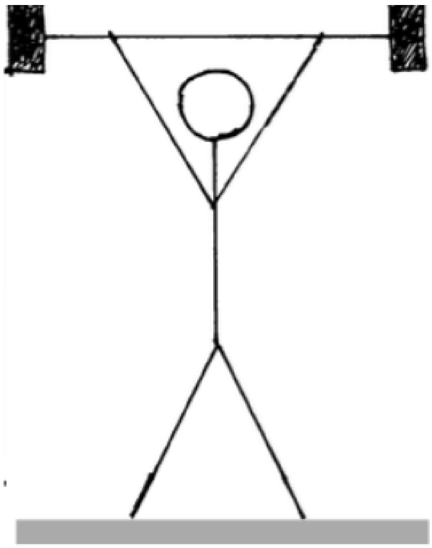
1000N

100N

188N

○ 288N

Question 13 4 pts



**Question: A weightlifter with a mass of 100 kg lifts a 100 kg barbell over his head as shown (a) D[illegible]**

(c) If the weightlifter holds the weight motionless over his head, what is the normal force at his feet?

from the ground he is standing on

○ about 1000N

○ about 100N

○ about 2000N

○ about 200N

Question 14 4 pts

As discussed in class. An astronaut is in the space station (stuck till february!) . We say he is weightless. What is true

- there is no more normal force to support him, astronaut does have weight
- the normal force acting on the astronaut balances out his weight inside the station
- I space out in class
- there is no more normal force to support him and gravity is 0 upthere

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